

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\localizer

TA: 0:19 PAT: Off Voxel size: 2.2×1.1×10.0 mm Rel. SNR: 1.00 SIEMENS: gre

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | On     |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | Off    |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 5                 |
| Dist. factor       | 20 %              |
| Position           | Isocenter         |
| Orientation        | Sagittal          |
| Phase enc. dir.    | A >> P            |
| Rotation           | 0.00 deg          |
| Slice group 2      |                   |
| Slices             | 1                 |
| Dist. factor       | 20 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | A >> P            |
| Rotation           | 0.00 deg          |
| Slice group 3      |                   |
| Slices             | 1                 |
| Dist. factor       | 20 %              |
| Position           | Isocenter         |
| Orientation        | Coronal           |
| Phase enc. dir.    | R >> L            |
| Rotation           | 0.00 deg          |
| Phase oversampling | 0 %               |
| FoV read           | 280 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 10.0 mm           |
| TR                 | 20.0 ms           |
| TE                 | 5.00 ms           |
| Averages           | 1                 |
| Concatenations     | 7                 |
| Filter             | Elliptical filter |
| Coil elements      | HEA;HEP           |

## Contrast

|                   |                  |
|-------------------|------------------|
| TD                | 0 ms             |
| MTC               | Off              |
| Magn. preparation | None             |
| Flip angle        | 40 deg           |
| Fat suppr.        | None             |
| Water suppr.      | None             |
| SWI               | Off              |
| Averaging mode    | Short term       |
| Reconstruction    | Magnitude        |
| Measurements      | 1                |
| Multiple series   | Each measurement |

## Resolution

|                 |     |
|-----------------|-----|
| Base resolution | 256 |
|-----------------|-----|

|                       |           |
|-----------------------|-----------|
| Phase resolution      | 50 %      |
| Phase partial Fourier | Off       |
| Interpolation         | On        |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Image Filter          | Off       |
| Distortion Corr.      | Off       |
| Prescan Normalize     | Off       |
| Normalize             | Off       |
| B1 filter             | Off       |
| Raw filter            | Off       |
| Elliptical filter     | On        |
| Mode                  | Inplane   |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Sequential  |
| Series           | Interleaved |
| Saturation mode  | Standard    |
| Special sat.     | None        |
| Tim CT mode      | Off         |

## System

|                          |                  |
|--------------------------|------------------|
| Body                     | Off              |
| HEP                      | On               |
| HEA                      | On               |
| Positioning mode         | REF              |
| Table position           | H                |
| Table position           | 0 mm             |
| MSMA                     | S - C - T        |
| Sagittal                 | L >> R           |
| Coronal                  | P >> A           |
| Transversal              | F >> H           |
| Save uncombined          | Off              |
| Coil Combine Mode        | Adaptive Combine |
| AutoAlign                | ---              |
| Auto Coil Select         | Default          |
| Shim mode                | Tune up          |
| Adjust with body coil    | Off              |
| Confirm freq. adjustment | Off              |
| Assume Silicone          | Off              |
| ? Ref. amplitude 1H      | 0.000 V          |
| Adjustment Tolerance     | Auto             |
| Adjust volume            |                  |
| Position                 | Isocenter        |
| Orientation              | Transversal      |
| Rotation                 | 0.00 deg         |
| R >> L                   | 350 mm           |
| A >> P                   | 263 mm           |
| F >> H                   | 350 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
| Segments        | 1    |
| Dark blood      | Off  |
| Resp. control   | Off  |

## Inline

|                    |     |
|--------------------|-----|
| Subtract           | Off |
| Liver registration | Off |
| Std-Dev-Sag        | Off |
| Std-Dev-Cor        | Off |

# SIEMENS MAGNETOM TrioTim syngo MR B17

|                      |     |
|----------------------|-----|
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |
| <hr/>                |     |
| Wash - In            | Off |
| Wash - Out           | Off |
| TTP                  | Off |
| PEI                  | Off |
| MIP - time           | Off |

## Sequence

|                     |            |
|---------------------|------------|
| Introduction        | On         |
| Dimension           | 2D         |
| Phase stabilisation | Off        |
| Asymmetric echo     | Off        |
| Contrasts           | 1          |
| Bandwidth           | 180 Hz/Px  |
| Flow comp.          | No         |
| Allowed delay       | 0 s        |
| <hr/>               |            |
| RF pulse type       | Fast       |
| Gradient mode       | Normal     |
| Excitation          | Slice-sel. |
| RF spoiling         | On         |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\AAHScout

TA: 0:17 PAT: 2 Voxel size: 1.6×1.6×1.6 mm Rel. SNR: 1.00 SIEMENS: AALScout

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | On     |
| Load images to graphic segments               | On     |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | Off    |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |           |
|--------------------|-----------|
| Slab group 1       |           |
| Slabs              | 1         |
| Dist. factor       | 20 %      |
| Position           | Isocenter |
| Orientation        | Sagittal  |
| Phase enc. dir.    | A >> P    |
| Rotation           | 0 deg     |
| AutoAlign          | Head      |
| Phase oversampling | 0 %       |
| Slice oversampling | 0.0 %     |
| Slices per slab    | 128       |
| FoV read           | 260 mm    |
| FoV phase          | 100.0 %   |
| Slice thickness    | 1.6 mm    |
| TR                 | 3.15 ms   |
| TE                 | 1.37 ms   |
| Averages           | 1         |
| Concatenations     | 1         |
| Filter             | Normalize |
| Coil elements      | HEA;HEP   |

## Contrast

|                |            |
|----------------|------------|
| Flip angle     | 8 deg      |
| Averaging mode | Short term |
| Reconstruction | Magnitude  |
| Measurements   | 1          |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 160           |
| Phase resolution      | 100 %         |
| Slice resolution      | 69 %          |
| Phase partial Fourier | 6/8           |
| Slice partial Fourier | 6/8           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Integrated    |
| Image Filter          | Off           |
| Distortion Corr.      | Off           |
| Prescan Normalize     | Off           |
| Normalize             | On            |
| Intensity             | Medium        |
| Cut off               | 20            |
| Width                 | 4             |
| Unfiltered images     | Off           |

|                   |     |
|-------------------|-----|
| B1 filter         | Off |
| Raw filter        | Off |
| Elliptical filter | Off |

## Geometry

|                  |            |
|------------------|------------|
| Multi-slice mode | Sequential |
| Series           | Ascending  |

## System

|                          |                  |
|--------------------------|------------------|
| Body                     | Off              |
| HEP                      | On               |
| HEA                      | On               |
| Positioning mode         | REF              |
| Table position           | H                |
| Table position           | 0 mm             |
| MSMA                     | S - C - T        |
| Sagittal                 | R >> L           |
| Coronal                  | A >> P           |
| Transversal              | F >> H           |
| Save uncombined          | Off              |
| Coil Combine Mode        | Adaptive Combine |
| Auto Coil Select         | Off              |
| Shim mode                | Tune up          |
| Adjust with body coil    | Off              |
| Confirm freq. adjustment | Off              |
| Assume Silicone          | Off              |
| ? Ref. amplitude 1H      | 0.000 V          |
| Adjustment Tolerance     | Auto             |
| Adjust volume            |                  |
| Position                 | Isocenter        |
| Orientation              | Transversal      |
| Rotation                 | 0.00 deg         |
| R >> L                   | 350 mm           |
| A >> P                   | 263 mm           |
| F >> H                   | 350 mm           |

## Inline

|                |       |
|----------------|-------|
| Time to center | 7.5 s |
|----------------|-------|

## Sequence

|                 |           |
|-----------------|-----------|
| Introduction    | On        |
| Dimension       | 3D        |
| Asymmetric echo | Weak      |
| Contrasts       | 1         |
| Bandwidth       | 550 Hz/Px |
| RF pulse type   | Fast      |
| Gradient mode   | Normal    |
| Excitation      | Non-sel.  |
| RF spoiling     | On        |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\me\_mprage

TA: 4:32 PAT: 2 Voxel size: 1.3×1.0×1.3 mm Rel. SNR: 1.00 USER: tfl\_mgh\_multiecho

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slab group 1       |                   |
| Slabs              | 1                 |
| Dist. factor       | 50 %              |
| Position           | R2.2 P1.7 F31.8   |
| Orientation        | Sagittal          |
| Phase enc. dir.    | A >> P            |
| Rotation           | 0.00 deg          |
| Phase oversampling | 0 %               |
| Slice oversampling | 0.0 %             |
| Slices per slab    | 128               |
| FoV read           | 256 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 1.33 mm           |
| TR                 | 2530 ms           |
| TE 1               | 3.31 ms           |
| TE 2               | 6.99 ms           |
| TE 3               | 8.85 ms           |
| TE 4               | 10.71 ms          |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                   |                  |
|-------------------|------------------|
| Magn. preparation | Non-sel. IR      |
| TI                | 1100 ms          |
| Flip angle        | 7.0 deg          |
| Fat suppr.        | None             |
| Water suppr.      | None             |
| Averaging mode    | Long term        |
| Reconstruction    | Magnitude        |
| Measurements      | 1                |
| Multiple series   | Each measurement |

## Resolution

|                       |       |
|-----------------------|-------|
| Base resolution       | 256   |
| Phase resolution      | 75 %  |
| Slice resolution      | 100 % |
| Phase partial Fourier | Off   |
| Slice partial Fourier | Off   |
| Interpolation         | Off   |

|                     |               |
|---------------------|---------------|
| PAT mode            | GRAPPA        |
| Accel. factor PE    | 2             |
| Ref. lines PE       | 24            |
| Matrix Coil Mode    | Auto (Triple) |
| Reference scan mode | Integrated    |

|                   |     |
|-------------------|-----|
| Image Filter      | Off |
| Distortion Corr.  | Off |
| Unfiltered images | Off |
| Prescan Normalize | On  |
| Normalize         | Off |
| B1 filter         | Off |
| Raw filter        | Off |
| Elliptical filter | Off |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Single shot |
| Series           | Interleaved |

## System

|                          |                  |
|--------------------------|------------------|
| Body                     | Off              |
| HEP                      | On               |
| HEA                      | On               |
| Positioning mode         | REF              |
| Table position           | H                |
| Table position           | 0 mm             |
| MSMA                     | S - C - T        |
| Sagittal                 | R >> L           |
| Coronal                  | A >> P           |
| Transversal              | F >> H           |
| Save uncombined          | Off              |
| Coil Combine Mode        | Adaptive Combine |
| AutoAlign                | Head > Brain     |
| Auto Coil Select         | Default          |
| Shim mode                | Standard         |
| Adjust with body coil    | Off              |
| Confirm freq. adjustment | Off              |
| Assume Silicone          | Off              |
| ? Ref. amplitude 1H      | 0.000 V          |
| Adjustment Tolerance     | Auto             |
| Adjust volume            |                  |
| Position                 | R2.2 P1.7 F31.8  |
| Orientation              | Sagittal         |
| Rotation                 | 0.00 deg         |
| F >> H                   | 256 mm           |
| A >> P                   | 256 mm           |
| R >> L                   | 171 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
| Dark blood      | Off  |

## Inline

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Std-Dev-Sag          | Off |
| Std-Dev-Cor          | Off |
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |

## Sequence

|                     |           |
|---------------------|-----------|
| Introduction        | On        |
| Dimension           | 3D        |
| Elliptical scanning | Off       |
| Asymmetric echo     | Off       |
| Contrasts           | 4         |
| Bandwidth 1         | 195 Hz/Px |

# SIEMENS MAGNETOM TrioTim syngo MR B17

|                        |           |
|------------------------|-----------|
| Bandwidth 2            | 651 Hz/Px |
| Bandwidth 3            | 651 Hz/Px |
| Bandwidth 4            | 651 Hz/Px |
| Flow comp. 1           | No        |
| Flow comp. 2           | No        |
| Flow comp. 3           | No        |
| Flow comp. 4           | No        |
| Echo spacing           | 13.8 ms   |
| <hr/>                  |           |
| RF pulse type          | Fast      |
| Gradient mode          | Fast      |
| Excitation             | Non-sel.  |
| RF spoiling            | On        |
| <hr/>                  |           |
| Readout polarity       | Positive  |
| Readout trajectory     | Bipolar   |
| Add. scale factor      | 8.0       |
| Gradient spoiling      | Integral  |
| Gradient moment factor | 3.0       |
| Siemens reconstruction | On        |
| Save raw k-space data  | Off       |
| Averaging              | RMS       |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\DOUBLE ECHO TSE low sar

TA: 2:02 PAT: Off Voxel size: 1.3×0.9×5.5 mm Rel. SNR: 1.00 SIEMENS: tse

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                    |                 |
|--------------------|-----------------|
| Slice group 1      |                 |
| Slices             | 20              |
| Dist. factor       | 40 %            |
| Position           | L7.1 A20.7 H7.3 |
| Orientation        | Transversal     |
| Phase enc. dir.    | R >> L          |
| Rotation           | 90.00 deg       |
| Phase oversampling | 0 %             |
| FoV read           | 240 mm          |
| FoV phase          | 75.0 %          |
| Slice thickness    | 5.5 mm          |
| TR                 | 4010 ms         |
| TE 1               | 33 ms           |
| TE 2               | 98 ms           |
| Averages           | 1               |
| Concatenations     | 1               |
| Filter             | None            |
| Coil elements      | HEA;HEP         |

## Contrast

|                   |            |
|-------------------|------------|
| MTC               | Off        |
| Magn. preparation | None       |
| Flip angle        | 160 deg    |
| Fat suppr.        | None       |
| Water suppr.      | None       |
| Restore magn.     | Off        |
| Averaging mode    | Short term |
| Reconstruction    | Magnitude  |
| Measurements      | 1          |
| Multiple series   | Off        |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 256       |
| Phase resolution      | 75 %      |
| Phase partial Fourier | Off       |
| Trajectory            | Cartesian |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Image Filter          | Off       |
| Distortion Corr.      | Off       |
| Prescan Normalize     | Off       |
| Normalize             | Off       |
| B1 filter             | Off       |
| Raw filter            | Off       |
| Elliptical filter     | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | Parallel F  |
| Gap              | 10 mm       |
| Thickness        | 50 mm       |
| Tim CT mode      | Off         |

## System

|                          |                  |
|--------------------------|------------------|
| Body                     | Off              |
| HEP                      | On               |
| HEA                      | On               |
| Positioning mode         | REF              |
| Table position           | H                |
| Table position           | 0 mm             |
| MSMA                     | S - C - T        |
| Sagittal                 | R >> L           |
| Coronal                  | A >> P           |
| Transversal              | F >> H           |
| Save uncombined          | Off              |
| Coil Combine Mode        | Adaptive Combine |
| AutoAlign                | ---              |
| Auto Coil Select         | Default          |
| Shim mode                | Tune up          |
| Adjust with body coil    | Off              |
| Confirm freq. adjustment | Off              |
| Assume Silicone          | Off              |
| ? Ref. amplitude 1H      | 0.000 V          |
| Adjustment Tolerance     | Auto             |
| Adjust volume            |                  |
| Position                 | Isocenter        |
| Orientation              | Transversal      |
| Rotation                 | 0.00 deg         |
| R >> L                   | 350 mm           |
| A >> P                   | 263 mm           |
| F >> H                   | 350 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
| Dark blood      | Off  |
| Resp. control   | Off  |

## Inline

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Std-Dev-Sag          | Off |
| Std-Dev-Cor          | Off |
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |

## Sequence

|                     |           |
|---------------------|-----------|
| Introduction        | On        |
| Dimension           | 2D        |
| Compensate T2 decay | Off       |
| Reduce Motion Sens. | Off       |
| Contrasts           | 2         |
| Bandwidth           | 179 Hz/Px |

## SIEMENS MAGNETOM TrioTim syngo MR B17

|                       |              |
|-----------------------|--------------|
| Flow comp.            | Read         |
| Allowed delay         | 0 s          |
| Echo spacing          | 10.9 ms      |
| <hr/>                 |              |
| Define                | Turbo factor |
| Turbo factor          | 5            |
| Echo trains per slice | 29           |
| RF pulse type         | Low SAR      |
| Gradient mode         | Fast*        |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\Flair axial low SAR

TA: 2:08 PAT: Off Voxel size: 1.3×0.9×5.5 mm Rel. SNR: 1.00 ! USER: tse\_crusher

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                    |                 |
|--------------------|-----------------|
| Slice group 1      |                 |
| Slices             | 20              |
| Dist. factor       | 40 %            |
| Position           | L7.1 A20.7 H7.3 |
| Orientation        | Transversal     |
| Phase enc. dir.    | R >> L          |
| Rotation           | 90.00 deg       |
| Phase oversampling | 0 %             |
| FoV read           | 240 mm          |
| FoV phase          | 75.0 %          |
| Slice thickness    | 5.5 mm          |
| TR                 | 9820 ms         |
| TE                 | 93 ms           |
| Averages           | 1               |
| Concatenations     | 1               |
| Filter             | Normalize       |
| Coil elements      | HEA;HEP         |

## Contrast

|                          |               |
|--------------------------|---------------|
| MTC                      | Off           |
| Magn. preparation        | Slice-sel. IR |
| TI                       | 2500 ms       |
| Freeze suppressed tissue | Off           |
| Flip angle               | 180 deg       |
| Fat suppr.               | None          |
| Water suppr.             | None          |
| Restore magn.            | Off           |
| Averaging mode           | Short term    |
| Reconstruction           | Magnitude     |
| Measurements             | 1             |
| Multiple series          | Off           |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 256       |
| Phase resolution      | 75 %      |
| Phase partial Fourier | Off       |
| Trajectory            | Cartesian |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Image Filter          | Off       |
| Distortion Corr.      | Off       |
| Prescan Normalize     | Off       |
| Normalize             | On        |
| Intensity             | Medium    |
| Cut off               | 20        |

|                   |     |
|-------------------|-----|
| Width             | 4   |
| Unfiltered images | Off |
| B1 filter         | Off |
| Raw filter        | Off |
| Elliptical filter | Off |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |
| Tim CT mode      | Off         |

## System

|                          |                  |
|--------------------------|------------------|
| Body                     | Off              |
| HEP                      | On               |
| HEA                      | On               |
| Positioning mode         | FIX              |
| Table position           | H                |
| Table position           | 0 mm             |
| MSMA                     | S - C - T        |
| Sagittal                 | R >> L           |
| Coronal                  | A >> P           |
| Transversal              | F >> H           |
| Save uncombined          | Off              |
| Coil Combine Mode        | Adaptive Combine |
| AutoAlign                | ---              |
| Auto Coil Select         | Default          |
| Shim mode                | Tune up          |
| Adjust with body coil    | Off              |
| Confirm freq. adjustment | Off              |
| Assume Silicone          | Off              |
| ? Ref. amplitude 1H      | 0.000 V          |
| Adjustment Tolerance     | Auto             |
| Adjust volume            |                  |
| Position                 | Isocenter        |
| Orientation              | Transversal      |
| Rotation                 | 0.00 deg         |
| R >> L                   | 350 mm           |
| A >> P                   | 263 mm           |
| F >> H                   | 350 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
| Dark blood      | Off  |
| Resp. control   | Off  |

## Inline

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Std-Dev-Sag          | Off |
| Std-Dev-Cor          | Off |
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |

## Sequence

|                     |     |
|---------------------|-----|
| Introduction        | Off |
| Dimension           | 2D  |
| Compensate T2 decay | Off |



## SIEMENS MAGNETOM TrioTim syngo MR B17

|                       |              |
|-----------------------|--------------|
| Reduce Motion Sens.   | Off          |
| Contrasts             | 1            |
| Bandwidth             | 199 Hz/Px    |
| Flow comp.            | Read         |
| Allowed delay         | 30 s         |
| Echo spacing          | 10.3 ms      |
| <hr/>                 |              |
| Define                | Turbo factor |
| Turbo factor          | 13           |
| Echo trains per slice | 12           |
| RF pulse type         | Low SAR      |
| Gradient mode         | Fast         |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\slice\_positioning

TA: 0:10 PAT: Off Voxel size: 3.4×3.4×3.5 mm Rel. SNR: 1.00 USER: cmrr\_mbep2d\_bold

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                          |                   |
|--------------------------|-------------------|
| Slice group 1            |                   |
| Slices                   | 48                |
| Dist. factor             | 0 %               |
| Position                 | L1.3 A0.6 F11.1   |
| Orientation              | Transversal       |
| Phase enc. dir.          | P >> A            |
| Rotation                 | -180.00 deg       |
| Phase oversampling       | 0 %               |
| FoV read                 | 220 mm            |
| FoV phase                | 100.0 %           |
| Slice thickness          | 3.50 mm           |
| TR                       | 400 ms            |
| TE                       | 30.0 ms           |
| Multi-band accel. factor | 16                |
| Filter                   | Prescan Normalize |
| Coil elements            | HEA;HEP           |

## Contrast

|                   |           |
|-------------------|-----------|
| MTC               | Off       |
| Magn. preparation | None      |
| Flip angle        | 24 deg    |
| Fat suppr.        | Fat sat.  |
| Averaging mode    | Long term |
| Reconstruction    | Magnitude |
| Measurements      | 1         |
| Delay in TR       | 0 ms      |
| Multiple series   | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 64        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 7/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | On        |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                 |
|--------------------------|-----------------|
| Body                     | Off             |
| HEP                      | On              |
| HEA                      | On              |
| Positioning mode         | FIX             |
| Table position           | H               |
| Table position           | 0 mm            |
| MSMA                     | S - C - T       |
| Sagittal                 | R >> L          |
| Coronal                  | A >> P          |
| Transversal              | F >> H          |
| Coil Combine Mode        | Sum of Squares  |
| AutoAlign                | Head > Brain    |
| Auto Coil Select         | Default         |
| Shim mode                | Standard        |
| Adjust with body coil    | Off             |
| Confirm freq. adjustment | Off             |
| Assume Silicone          | Off             |
| ? Ref. amplitude 1H      | 0.000 V         |
| Adjustment Tolerance     | Auto            |
| Adjust volume            |                 |
| Position                 | L1.3 A0.6 F11.1 |
| Orientation              | Transversal     |
| Rotation                 | -180.00 deg     |
| R >> L                   | 220 mm          |
| A >> P                   | 220 mm          |
| F >> H                   | 168 mm          |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## BOLD

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Dynamic t-maps          | Off      |
| Starting ignore meas    | 0        |
| Ignore after transition | 0        |
| Model transition states | On       |
| Temp. highpass filter   | On       |
| Threshold               | 4.00     |
| Paradigm size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion correction       | Off      |
| Spatial filter          | Off      |

## Sequence

# SIEMENS MAGNETOM TrioTim syngo MR B17

|                          |            |
|--------------------------|------------|
| Introduction             | Off        |
| Bandwidth                | 3720 Hz/Px |
| Echo spacing             | 0.5 ms     |
| <hr/>                    |            |
| EPI factor               | 64         |
| Gradient mode            | Fast       |
| <hr/>                    |            |
| Excite pulse duration    | 4000 us    |
| Single-band images       | Off        |
| MB RF phase scramble     | Off        |
| SENSE1 coil combine      | Off        |
| Log physiology to file   | Off        |
| Invert RO/PE polarity    | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Triggering scheme        | Standard   |
| Starting ignore meas     | 0          |
| Paradigm size            | 2          |
| Multiplier               | 1          |
| Step [1]                 | 1          |
| Step [2]                 | 0          |

## SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\ep2d\_se\_VASO

TA: 0:24 PAT: Off Voxel size: 3.0×3.0×5.0 mm Rel. SNR: 1.00 USER: ep2d\_se\_VASO

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 1                 |
| Dist. factor       | 50 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 240 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 2000 ms           |
| TE                 | 24 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                   |             |
|-------------------|-------------|
| MTC               | Off         |
| Magn. preparation | Non-sel. IR |
| TI                | 711 ms      |
| Fat suppr.        | Fat sat.    |
| Averaging mode    | Long term   |
| Reconstruction    | Magnitude   |
| Measurements      | 10          |
| Delay in TR       | 0 ms        |
| Multiple series   | Off         |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 80        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 6/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | Off       |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |

## Special sat.

None

## System

|                   |                |
|-------------------|----------------|
| Body              | Off            |
| HEP               | On             |
| HEA               | On             |
| Positioning mode  | REF            |
| Table position    | H              |
| Table position    | 0 mm           |
| MSMA              | S - C - T      |
| Sagittal          | R >> L         |
| Coronal           | A >> P         |
| Transversal       | F >> H         |
| Save uncombined   | Off            |
| Coil Combine Mode | Sum of Squares |
| AutoAlign         | ---            |
| Auto Coil Select  | Default        |

|                          |             |
|--------------------------|-------------|
| Shim mode                | Standard    |
| Adjust with body coil    | Off         |
| Confirm freq. adjustment | Off         |
| Assume Silicone          | Off         |
| ? Ref. amplitude 1H      | 0.000 V     |
| Adjustment Tolerance     | Auto        |
| Adjust volume            |             |
| Position                 | Isocenter   |
| Orientation              | Transversal |
| Rotation                 | 180.00 deg  |
| R >> L                   | 240 mm      |
| A >> P                   | 240 mm      |
| F >> H                   | 5 mm        |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## Inline

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Std-Dev-Sag          | Off |
| Std-Dev-Cor          | Off |
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |

## Sequence

|                   |            |
|-------------------|------------|
| Introduction      | Off        |
| Bandwidth         | 2604 Hz/Px |
| Free echo spacing | Off        |
| Echo spacing      | 0.47 ms    |
| EPI factor        | 80         |
| RF pulse type     | Normal     |
| Gradient mode     | Fast       |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\VASO\_flipback\_464\_550

TA: 0:47 PAT: Off Voxel size: 3.0×3.0×5.0 mm Rel. SNR: 1.00 USER: mjd\_mclean\_flipback

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 1                 |
| Dist. factor       | 50 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 240 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 550 ms            |
| TE                 | 24 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                   |           |
|-------------------|-----------|
| MTC               | Off       |
| Magn. preparation | None      |
| Fat suppr.        | Fat sat.  |
| Averaging mode    | Long term |
| Reconstruction    | Magnitude |
| Measurements      | 80        |
| Delay in TR       | 0 ms      |
| Multiple series   | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 80        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 6/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | Off       |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 240 mm         |
| A >> P                   | 240 mm         |
| F >> H                   | 5 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## Inline

|                      |     |
|----------------------|-----|
| Subtract             | On  |
| Autoscaling          | Off |
| Scaling factor       | 1   |
| Offset               | 0   |
| Subtrahend           | 1   |
| Std-Dev-Sag          | Off |
| Std-Dev-Cor          | Off |
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |

## Sequence

|                            |            |
|----------------------------|------------|
| Introduction               | Off        |
| Bandwidth                  | 2604 Hz/Px |
| Free echo spacing          | Off        |
| Echo spacing               | 0.47 ms    |
| EPI factor                 | 80         |
| RF pulse type              | Normal     |
| Gradient mode              | Fast       |
| Inversion time             | 464 ms     |
| Inversion time             | 464 ms     |
| Sli-Sel Gradient Direction | 0 s0;p1    |
| Inversion Pulse            | 0 RF num   |
| HS Flip Angle              | 2000 deg   |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\VASO\_flipback\_614\_788

TA: 0:50 PAT: Off Voxel size: 3.0×3.0×5.0 mm Rel. SNR: 1.00 USER: mjd\_mclean\_flipback

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 1                 |
| Dist. factor       | 50 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 240 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 788 ms            |
| TE                 | 24 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                   |           |
|-------------------|-----------|
| MTC               | Off       |
| Magn. preparation | None      |
| Fat suppr.        | Fat sat.  |
| Averaging mode    | Long term |
| Reconstruction    | Magnitude |
| Measurements      | 60        |
| Delay in TR       | 0 ms      |
| Multiple series   | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 80        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 6/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | Off       |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 240 mm         |
| A >> P                   | 240 mm         |
| F >> H                   | 5 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## Inline

|                      |     |
|----------------------|-----|
| Subtract             | On  |
| Autoscaling          | Off |
| Scaling factor       | 1   |
| Offset               | 0   |
| Subtrahend           | 1   |
| Std-Dev-Sag          | Off |
| Std-Dev-Cor          | Off |
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |

## Sequence

|                            |            |
|----------------------------|------------|
| Introduction               | Off        |
| Bandwidth                  | 2604 Hz/Px |
| Free echo spacing          | Off        |
| Echo spacing               | 0.47 ms    |
| EPI factor                 | 80         |
| RF pulse type              | Normal     |
| Gradient mode              | Fast       |
| Inversion time             | 614 ms     |
| Inversion time             | 614 ms     |
| Sli-Sel Gradient Direction | 0 s0;p1    |
| Inversion Pulse            | 0 RF num   |
| HS Flip Angle              | 2000 deg   |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\VASO\_flipback\_764\_1085

TA: 0:47 PAT: Off Voxel size: 3.0×3.0×5.0 mm Rel. SNR: 1.00 USER: mjd\_mclean\_flipback

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 1                 |
| Dist. factor       | 50 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 240 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 1090 ms           |
| TE                 | 24 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                   |           |
|-------------------|-----------|
| MTC               | Off       |
| Magn. preparation | None      |
| Fat suppr.        | Fat sat.  |
| Averaging mode    | Long term |
| Reconstruction    | Magnitude |
| Measurements      | 40        |
| Delay in TR       | 0 ms      |
| Multiple series   | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 80        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 6/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | Off       |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 240 mm         |
| A >> P                   | 240 mm         |
| F >> H                   | 5 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## Inline

|                      |     |
|----------------------|-----|
| Subtract             | On  |
| Autoscaling          | Off |
| Scaling factor       | 1   |
| Offset               | 0   |
| Subtrahend           | 1   |
| Std-Dev-Sag          | Off |
| Std-Dev-Cor          | Off |
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |

## Sequence

|                            |            |
|----------------------------|------------|
| Introduction               | Off        |
| Bandwidth                  | 2604 Hz/Px |
| Free echo spacing          | Off        |
| Echo spacing               | 0.47 ms    |
| EPI factor                 | 80         |
| RF pulse type              | Normal     |
| Gradient mode              | Fast       |
| Inversion time             | 764 ms     |
| Inversion time             | 764 ms     |
| Sli-Sel Gradient Direction | 0 s0;p1    |
| Inversion Pulse            | 0 RF num   |
| HS Flip Angle              | 2000 deg   |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\VASO\_flipback\_914\_1492

TA: 0:49 PAT: Off Voxel size: 3.0×3.0×5.0 mm Rel. SNR: 1.00 USER: mjd\_mclean\_flipback

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 1                 |
| Dist. factor       | 50 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 240 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 1490 ms           |
| TE                 | 24 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                   |           |
|-------------------|-----------|
| MTC               | Off       |
| Magn. preparation | None      |
| Fat suppr.        | Fat sat.  |
| Averaging mode    | Long term |
| Reconstruction    | Magnitude |
| Measurements      | 30        |
| Delay in TR       | 0 ms      |
| Multiple series   | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 80        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 6/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | Off       |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 240 mm         |
| A >> P                   | 240 mm         |
| F >> H                   | 5 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## Inline

|                      |     |
|----------------------|-----|
| Subtract             | On  |
| Autoscaling          | Off |
| Scaling factor       | 1   |
| Offset               | 0   |
| Subtrahend           | 1   |
| Std-Dev-Sag          | Off |
| Std-Dev-Cor          | Off |
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |

## Sequence

|                            |            |
|----------------------------|------------|
| Introduction               | Off        |
| Bandwidth                  | 2604 Hz/Px |
| Free echo spacing          | Off        |
| Echo spacing               | 0.47 ms    |
| EPI factor                 | 80         |
| RF pulse type              | Normal     |
| Gradient mode              | Fast       |
| Inversion time             | 914 ms     |
| Inversion time             | 914 ms     |
| Sli-Sel Gradient Direction | 0 s0;p1    |
| Inversion Pulse            | 0 RF num   |
| HS Flip Angle              | 2000 deg   |



# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\VASO\_flipback\_1066\_2190

TA: 0:48 PAT: Off Voxel size: 3.0×3.0×5.0 mm Rel. SNR: 1.00 USER: mjd\_mclean\_flipback

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 1                 |
| Dist. factor       | 50 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 240 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 2190 ms           |
| TE                 | 24 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                   |           |
|-------------------|-----------|
| MTC               | Off       |
| Magn. preparation | None      |
| Fat suppr.        | Fat sat.  |
| Averaging mode    | Long term |
| Reconstruction    | Magnitude |
| Measurements      | 20        |
| Delay in TR       | 0 ms      |
| Multiple series   | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 80        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 6/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | Off       |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 240 mm         |
| A >> P                   | 240 mm         |
| F >> H                   | 5 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## Inline

|                      |     |
|----------------------|-----|
| Subtract             | On  |
| Autoscaling          | Off |
| Scaling factor       | 1   |
| Offset               | 0   |
| Subtrahend           | 1   |
| Std-Dev-Sag          | Off |
| Std-Dev-Cor          | Off |
| Std-Dev-Tra          | Off |
| Std-Dev-Time         | Off |
| MIP-Sag              | Off |
| MIP-Cor              | Off |
| MIP-Tra              | Off |
| MIP-Time             | Off |
| Save original images | On  |

## Sequence

|                            |            |
|----------------------------|------------|
| Introduction               | Off        |
| Bandwidth                  | 2604 Hz/Px |
| Free echo spacing          | Off        |
| Echo spacing               | 0.47 ms    |
| EPI factor                 | 80         |
| RF pulse type              | Normal     |
| Gradient mode              | Fast       |
| Inversion time             | 1066 ms    |
| Inversion time             | 1066 ms    |
| Sli-Sel Gradient Direction | 0 s0;p1    |
| Inversion Pulse            | 0 RF num   |
| HS Flip Angle              | 2000 deg   |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\restingstate

TA: 6:10 PAT: Off Voxel size: 3.4×3.4×3.5 mm Rel. SNR: 1.00 USER: cmrr\_mbep2d\_bold

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                          |                   |
|--------------------------|-------------------|
| Slice group 1            |                   |
| Slices                   | 48                |
| Dist. factor             | 0 %               |
| Position                 | L1.3 A0.6 F11.1   |
| Orientation              | Transversal       |
| Phase enc. dir.          | P >> A            |
| Rotation                 | -180.00 deg       |
| Phase oversampling       | 0 %               |
| FoV read                 | 220 mm            |
| FoV phase                | 100.0 %           |
| Slice thickness          | 3.50 mm           |
| TR                       | 400 ms            |
| TE                       | 30.0 ms           |
| Multi-band accel. factor | 16                |
| Filter                   | Prescan Normalize |
| Coil elements            | HEA;HEP           |

## Contrast

|                   |           |
|-------------------|-----------|
| MTC               | Off       |
| Magn. preparation | None      |
| Flip angle        | 24 deg    |
| Fat suppr.        | Fat sat.  |
| Averaging mode    | Long term |
| Reconstruction    | Magnitude |
| Measurements      | 900       |
| Delay in TR       | 0 ms      |
| Multiple series   | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 64        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 7/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | On        |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                 |
|--------------------------|-----------------|
| Body                     | Off             |
| HEP                      | On              |
| HEA                      | On              |
| Positioning mode         | FIX             |
| Table position           | H               |
| Table position           | 0 mm            |
| MSMA                     | S - C - T       |
| Sagittal                 | R >> L          |
| Coronal                  | A >> P          |
| Transversal              | F >> H          |
| Coil Combine Mode        | Sum of Squares  |
| AutoAlign                | Head > Brain    |
| Auto Coil Select         | Default         |
| Shim mode                | Standard        |
| Adjust with body coil    | Off             |
| Confirm freq. adjustment | Off             |
| Assume Silicone          | Off             |
| ? Ref. amplitude 1H      | 0.000 V         |
| Adjustment Tolerance     | Auto            |
| Adjust volume            |                 |
| Position                 | L1.3 A0.6 F11.1 |
| Orientation              | Transversal     |
| Rotation                 | -180.00 deg     |
| R >> L                   | 220 mm          |
| A >> P                   | 220 mm          |
| F >> H                   | 168 mm          |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## BOLD

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Dynamic t-maps          | Off      |
| Starting ignore meas    | 0        |
| Ignore after transition | 0        |
| Model transition states | On       |
| Temp. highpass filter   | On       |
| Threshold               | 4.00     |
| Paradigm size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion correction       | Off      |
| Spatial filter          | Off      |

## Sequence

# SIEMENS MAGNETOM TrioTim syngo MR B17

|                          |            |
|--------------------------|------------|
| Introduction             | Off        |
| Bandwidth                | 3720 Hz/Px |
| Echo spacing             | 0.5 ms     |
| <hr/>                    |            |
| EPI factor               | 64         |
| Gradient mode            | Fast       |
| <hr/>                    |            |
| Excite pulse duration    | 4000 us    |
| Single-band images       | Off        |
| MB RF phase scramble     | Off        |
| SENSE1 coil combine      | Off        |
| Log physiology to file   | Off        |
| Invert RO/PE polarity    | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Triggering scheme        | Standard   |
| Starting ignore meas     | 0          |
| Paradigm size            | 2          |
| Multiplier               | 1          |
| Step [1]                 | 1          |
| Step [2]                 | 0          |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\pcasl\_DE\_M0

TA: 0:40 PAT: 2 Voxel size: 3.4×3.4×5.0 mm Rel. SNR: 1.00 USER: ep2d\_DE\_pcasl\_iPAT

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 19                |
| Dist. factor       | 40 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 220 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 10000 ms          |
| TE 1               | 10.0 ms           |
| TE 2               | 25 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 90 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 1         |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 64            |
| Phase resolution      | 100 %         |
| Phase partial Fourier | 7/8           |
| Interpolation         | Off           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Separate      |
| Distortion Corr.      | Off           |
| Unfiltered images     | Off           |
| Prescan Normalize     | On            |
| Raw filter            | Off           |
| Elliptical filter     | Off           |
| Hamming               | Off           |

## Geometry

Multi-slice mode  
Series Interleaved  
Ascending

Special sat. None

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 220 mm         |
| A >> P                   | 220 mm         |
| F >> H                   | 131 mm         |

## Physio

1st Signal/Mode None

## Perf

|                 |     |
|-----------------|-----|
| GBP             | Off |
| PBP             | Off |
| TTP             | Off |
| Original images | On  |

## Sequence

|                   |             |
|-------------------|-------------|
| Introduction      | Off         |
| Contrasts         | 2           |
| Bandwidth         | 3004 Hz/Px  |
| Free echo spacing | Off         |
| Echo spacing      | 0.44 ms     |
| EPI factor        | 64          |
| RF pulse type     | Normal      |
| Gradient mode     | Fast        |
| CASL Method       | Multi-slice |
| Label Offset      | 90 mm       |
| Post Label Delay  | 500000 us   |
| Num RF Blocks     | 82          |
| RF GAP            | 360 us      |
| mean Gz x10       | 6 mT/m      |
| phi adjust        | 100 percent |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\pcasl\_DE\_B0

TA: 0:21

PAT: Off

Voxel size: 3.4×3.4×5.0 mm

Rel. SNR: 1.00

USER: ep2d\_DE\_pcasl\_iPAT

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |             |
|--------------------|-------------|
| Slice group 1      |             |
| Slices             | 19          |
| Dist. factor       | 40 %        |
| Position           | Isocenter   |
| Orientation        | Transversal |
| Phase enc. dir.    | P >> A      |
| Rotation           | 180.00 deg  |
| Phase oversampling | 0 %         |
| FoV read           | 220 mm      |
| FoV phase          | 100.0 %     |
| Slice thickness    | 5.0 mm      |
| TR                 | 3500 ms     |
| TE                 | 15 ms       |
| Averages           | 1           |
| Concatenations     | 1           |
| Filter             | None        |
| Coil elements      | HEA;HEP     |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 90 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 4         |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 64        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 7/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Prescan Normalize     | Off       |
| Raw filter            | Off       |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Ascending   |
| Special sat.     | None        |

## System

|                   |                |
|-------------------|----------------|
| Body              | Off            |
| HEP               | On             |
| HEA               | On             |
| Positioning mode  | FIX            |
| Table position    | H              |
| Table position    | 0 mm           |
| MSMA              | S - C - T      |
| Sagittal          | R >> L         |
| Coronal           | A >> P         |
| Transversal       | F >> H         |
| Save uncombined   | Off            |
| Coil Combine Mode | Sum of Squares |
| AutoAlign         | ---            |
| Auto Coil Select  | Default        |

|                          |             |
|--------------------------|-------------|
| Shim mode                | Standard    |
| Adjust with body coil    | Off         |
| Confirm freq. adjustment | Off         |
| Assume Silicone          | Off         |
| ? Ref. amplitude 1H      | 0.000 V     |
| Adjustment Tolerance     | Auto        |
| Adjust volume            |             |
| Position                 | Isocenter   |
| Orientation              | Transversal |
| Rotation                 | 180.00 deg  |
| R >> L                   | 220 mm      |
| A >> P                   | 220 mm      |
| F >> H                   | 131 mm      |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## Perf

|                 |     |
|-----------------|-----|
| GBP             | Off |
| PBP             | Off |
| TTP             | Off |
| Original images | On  |

## Sequence

|                   |             |
|-------------------|-------------|
| Introduction      | Off         |
| Contrasts         | 1           |
| Bandwidth         | 3004 Hz/Px  |
| Free echo spacing | Off         |
| Echo spacing      | 0.44 ms     |
| EPI factor        | 64          |
| RF pulse type     | Normal      |
| Gradient mode     | Fast        |
| CASL Method       | Multi-slice |
| Label Offset      | 90 mm       |
| Post Label Delay  | 500000 us   |
| Num RF Blocks     | 82          |
| RF GAP            | 360 us      |
| mean Gz x10       | 6 mT/m      |
| phi adjust        | 100 percent |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\pcasl\_DE\_0p5

TA: 1:00 PAT: 2 Voxel size: 3.4×3.4×5.0 mm Rel. SNR: 1.00 USER: ep2d\_DE\_pcasl\_iPAT

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 19                |
| Dist. factor       | 40 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 220 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 3000 ms           |
| TE 1               | 10.0 ms           |
| TE 2               | 25 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 90 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 16        |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 64            |
| Phase resolution      | 100 %         |
| Phase partial Fourier | 7/8           |
| Interpolation         | Off           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Separate      |
| Distortion Corr.      | Off           |
| Unfiltered images     | Off           |
| Prescan Normalize     | On            |
| Raw filter            | Off           |
| Elliptical filter     | Off           |
| Hamming               | Off           |

## Geometry

Multi-slice mode  
Series

Interleaved  
Ascending

Special sat.

None

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 220 mm         |
| A >> P                   | 220 mm         |
| F >> H                   | 131 mm         |

## Physio

1st Signal/Mode

None

## Perf

|                 |     |
|-----------------|-----|
| GBP             | Off |
| PBP             | Off |
| TTP             | Off |
| Original images | On  |

## Sequence

|                   |             |
|-------------------|-------------|
| Introduction      | Off         |
| Contrasts         | 2           |
| Bandwidth         | 3004 Hz/Px  |
| Free echo spacing | Off         |
| Echo spacing      | 0.44 ms     |
| EPI factor        | 64          |
| RF pulse type     | Normal      |
| Gradient mode     | Fast        |
| CASL Method       | Multi-slice |
| Label Offset      | 90 mm       |
| Post Label Delay  | 500000 us   |
| Num RF Blocks     | 82          |
| RF GAP            | 360 us      |
| mean Gz x10       | 6 mT/m      |
| phi adjust        | 100 percent |

## SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\pcasl\_DE\_1p0

TA: 1:07 PAT: 2 Voxel size: 3.4×3.4×5.0 mm Rel. SNR: 1.00 USER: ep2d\_DE\_pcasl\_iPAT

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 19                |
| Dist. factor       | 40 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 220 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 3500 ms           |
| TE 1               | 10.0 ms           |
| TE 2               | 25 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 90 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 16        |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 64            |
| Phase resolution      | 100 %         |
| Phase partial Fourier | 7/8           |
| Interpolation         | Off           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Separate      |
| Distortion Corr.      | Off           |
| Unfiltered images     | Off           |
| Prescan Normalize     | On            |
| Raw filter            | Off           |
| Elliptical filter     | Off           |
| Hamming               | Off           |

## Geometry

Multi-slice mode  
Series Interleaved  
Ascending

Special sat. None

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 220 mm         |
| A >> P                   | 220 mm         |
| F >> H                   | 131 mm         |

## Physio

1st Signal/Mode None

## Perf

|                 |     |
|-----------------|-----|
| GBP             | Off |
| PBP             | Off |
| TTP             | Off |
| Original images | On  |

## Sequence

|                   |             |
|-------------------|-------------|
| Introduction      | Off         |
| Contrasts         | 2           |
| Bandwidth         | 3004 Hz/Px  |
| Free echo spacing | Off         |
| Echo spacing      | 0.44 ms     |
| EPI factor        | 64          |
| RF pulse type     | Normal      |
| Gradient mode     | Fast        |
| CASL Method       | Multi-slice |
| Label Offset      | 90 mm       |
| Post Label Delay  | 1000000 us  |
| Num RF Blocks     | 82          |
| RF GAP            | 360 us      |
| mean Gz x10       | 6 mT/m      |
| phi adjust        | 100 percent |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\pcasl\_DE\_1p5

TA: 1:16 PAT: 2 Voxel size: 3.4×3.4×5.0 mm Rel. SNR: 1.00 USER: ep2d\_DE\_pcasl\_iPAT

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 19                |
| Dist. factor       | 40 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 220 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 4000 ms           |
| TE 1               | 10.0 ms           |
| TE 2               | 25 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 90 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 16        |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 64            |
| Phase resolution      | 100 %         |
| Phase partial Fourier | 7/8           |
| Interpolation         | Off           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Separate      |
| Distortion Corr.      | Off           |
| Unfiltered images     | Off           |
| Prescan Normalize     | On            |
| Raw filter            | Off           |
| Elliptical filter     | Off           |
| Hamming               | Off           |

## Geometry

Multi-slice mode  
Series Interleaved  
Ascending

Special sat. None

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 220 mm         |
| A >> P                   | 220 mm         |
| F >> H                   | 131 mm         |

## Physio

1st Signal/Mode None

## Perf

|                 |     |
|-----------------|-----|
| GBP             | Off |
| PBP             | Off |
| TTP             | Off |
| Original images | On  |

## Sequence

|                   |             |
|-------------------|-------------|
| Introduction      | Off         |
| Contrasts         | 2           |
| Bandwidth         | 3004 Hz/Px  |
| Free echo spacing | Off         |
| Echo spacing      | 0.44 ms     |
| EPI factor        | 64          |
| RF pulse type     | Normal      |
| Gradient mode     | Fast        |
| CASL Method       | Multi-slice |
| Label Offset      | 90 mm       |
| Post Label Delay  | 1500000 us  |
| Num RF Blocks     | 82          |
| RF GAP            | 360 us      |
| mean Gz x10       | 6 mT/m      |
| phi adjust        | 100 percent |



## SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\pcasl\_DE\_2p0

TA: 1:26 PAT: 2 Voxel size: 3.4×3.4×5.0 mm Rel. SNR: 1.00 USER: ep2d\_DE\_pcasl\_iPAT

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 19                |
| Dist. factor       | 40 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 220 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 4500 ms           |
| TE 1               | 10.0 ms           |
| TE 2               | 25 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 90 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 16        |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 64            |
| Phase resolution      | 100 %         |
| Phase partial Fourier | 7/8           |
| Interpolation         | Off           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Separate      |
| Distortion Corr.      | Off           |
| Unfiltered images     | Off           |
| Prescan Normalize     | On            |
| Raw filter            | Off           |
| Elliptical filter     | Off           |
| Hamming               | Off           |

## Geometry

Multi-slice mode  
Series Interleaved  
Ascending

Special sat. None

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 220 mm         |
| A >> P                   | 220 mm         |
| F >> H                   | 131 mm         |

## Physio

1st Signal/Mode None

## Perf

|                 |     |
|-----------------|-----|
| GBP             | Off |
| PBP             | Off |
| TTP             | Off |
| Original images | On  |

## Sequence

|                   |             |
|-------------------|-------------|
| Introduction      | Off         |
| Contrasts         | 2           |
| Bandwidth         | 3004 Hz/Px  |
| Free echo spacing | Off         |
| Echo spacing      | 0.44 ms     |
| EPI factor        | 64          |
| RF pulse type     | Normal      |
| Gradient mode     | Fast        |
| CASL Method       | Multi-slice |
| Label Offset      | 90 mm       |
| Post Label Delay  | 2000000 us  |
| Num RF Blocks     | 82          |
| RF GAP            | 360 us      |
| mean Gz x10       | 6 mT/m      |
| phi adjust        | 100 percent |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\pcasl\_DE\_2p5

TA: 1:35 PAT: 2 Voxel size: 3.4×3.4×5.0 mm Rel. SNR: 1.00 USER: ep2d\_DE\_pcasl\_iPAT

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 19                |
| Dist. factor       | 40 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 220 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 5000 ms           |
| TE 1               | 10.0 ms           |
| TE 2               | 25 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 90 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 16        |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 64            |
| Phase resolution      | 100 %         |
| Phase partial Fourier | 7/8           |
| Interpolation         | Off           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Separate      |
| Distortion Corr.      | Off           |
| Unfiltered images     | Off           |
| Prescan Normalize     | On            |
| Raw filter            | Off           |
| Elliptical filter     | Off           |
| Hamming               | Off           |

## Geometry

Multi-slice mode  
Series Interleaved  
Ascending

Special sat. None

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 220 mm         |
| A >> P                   | 220 mm         |
| F >> H                   | 131 mm         |

## Physio

1st Signal/Mode None

## Perf

|                 |     |
|-----------------|-----|
| GBP             | Off |
| PBP             | Off |
| TTP             | Off |
| Original images | On  |

## Sequence

|                   |             |
|-------------------|-------------|
| Introduction      | Off         |
| Contrasts         | 2           |
| Bandwidth         | 3004 Hz/Px  |
| Free echo spacing | Off         |
| Echo spacing      | 0.44 ms     |
| EPI factor        | 64          |
| RF pulse type     | Normal      |
| Gradient mode     | Fast        |
| CASL Method       | Multi-slice |
| Label Offset      | 90 mm       |
| Post Label Delay  | 2500000 us  |
| Num RF Blocks     | 82          |
| RF GAP            | 360 us      |
| mean Gz x10       | 6 mT/m      |
| phi adjust        | 100 percent |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\dsc\_mri

TA: 3:09 PAT: 3 Voxel size: 1.8×1.8×3.5 mm Rel. SNR: 1.00 SIEMENS: ep2d\_pace

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | Off    |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 29                |
| Dist. factor       | 14 %              |
| Position           | L0.0 P10.9 H20.5  |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | -180.00 deg       |
| Phase oversampling | 0 %               |
| FoV read           | 224 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 3.5 mm            |
| TR                 | 1510 ms           |
| TE                 | 21 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 60 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 120       |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 128           |
| Phase resolution      | 100 %         |
| Phase partial Fourier | Off           |
| Interpolation         | Off           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 3             |
| Ref. lines PE         | 36            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Separate      |
| Distortion Corr.      | Off           |
| Unfiltered images     | Off           |
| Prescan Normalize     | On            |
| Raw filter            | On            |
| Elliptical filter     | Off           |
| Hamming               | Off           |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
|------------------|-------------|

## Series

## Ascending

|              |      |
|--------------|------|
| Special sat. | None |
|--------------|------|

## System

|                          |                  |
|--------------------------|------------------|
| Body                     | Off              |
| HEP                      | On               |
| HEA                      | On               |
| Positioning mode         | FIX              |
| Table position           | H                |
| Table position           | 0 mm             |
| MSMA                     | S - C - T        |
| Sagittal                 | R >> L           |
| Coronal                  | A >> P           |
| Transversal              | F >> H           |
| Coil Combine Mode        | Sum of Squares   |
| AutoAlign                | Head > Brain     |
| Auto Coil Select         | Default          |
| Shim mode                | Standard         |
| Adjust with body coil    | Off              |
| Confirm freq. adjustment | Off              |
| Assume Silicone          | Off              |
| ? Ref. amplitude 1H      | 0.000 V          |
| Adjustment Tolerance     | Auto             |
| Adjust volume            |                  |
| Position                 | L0.0 P10.9 H20.5 |
| Orientation              | Transversal      |
| Rotation                 | -180.00 deg      |
| R >> L                   | 224 mm           |
| A >> P                   | 224 mm           |
| F >> H                   | 116 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## BOLD

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Dynamic t-maps          | Off      |
| Starting ignore meas    | 0        |
| Ignore after transition | 0        |
| Model transition states | On       |
| Temp. highpass filter   | On       |
| Threshold               | 4.00     |
| Paradigm size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion correction       | On       |

## SIEMENS MAGNETOM TrioTim syngo MR B17

|                |            |
|----------------|------------|
| Interpolation  | 3D-K-space |
| Spatial filter | Off        |

### Sequence

|                   |            |
|-------------------|------------|
| Introduction      | Off        |
| Bandwidth         | 1698 Hz/Px |
| Free echo spacing | Off        |
| Echo spacing      | 0.69 ms    |
| <hr/>             |            |
| EPI factor        | 128        |
| RF pulse type     | Normal     |
| Gradient mode     | Fast       |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\gre\_field\_mapping

TA: 2:12

Voxel size: 3.4×1.7×3.5 mm

Rel. SNR: 1.00

SIEMENS: gre\_field\_mapping

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                    |                  |
|--------------------|------------------|
| Slice group 1      |                  |
| Slices             | 32               |
| Dist. factor       | 14 %             |
| Position           | L0.0 P10.9 H20.5 |
| Orientation        | Transversal      |
| Phase enc. dir.    | P >> A           |
| Rotation           | -180.00 deg      |
| Phase oversampling | 0 %              |
| FoV read           | 220 mm           |
| FoV phase          | 100.0 %          |
| Slice thickness    | 3.5 mm           |
| TR                 | 1000 ms          |
| TE 1               | 10.00 ms         |
| TE 2               | 12.46 ms         |
| Averages           | 1                |
| Concatenations     | 1                |
| Filter             | None             |
| Coil elements      | HEA;HEP          |

## Contrast

|                 |             |
|-----------------|-------------|
| MTC             | Off         |
| Flip angle      | 90 deg      |
| Fat suppr.      | None        |
| Averaging mode  | Short term  |
| Reconstruction  | Magn./Phase |
| Measurements    | 1           |
| Multiple series | Off         |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 128       |
| Phase resolution      | 50 %      |
| Phase partial Fourier | Off       |
| Interpolation         | Off       |
| Matrix Coil Mode      | Auto (CP) |
| Image Filter          | Off       |
| Distortion Corr.      | Off       |
| Prescan Normalize     | Off       |
| Normalize             | Off       |
| B1 filter             | Off       |
| Raw filter            | Off       |
| Elliptical filter     | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                  |
|--------------------------|------------------|
| Body                     | Off              |
| HEP                      | On               |
| HEA                      | On               |
| Positioning mode         | FIX              |
| Table position           | H                |
| Table position           | 0 mm             |
| MSMA                     | S - C - T        |
| Sagittal                 | R >> L           |
| Coronal                  | A >> P           |
| Transversal              | F >> H           |
| Save uncombined          | Off              |
| Coil Combine Mode        | Adaptive Combine |
| AutoAlign                | Head > Brain     |
| Auto Coil Select         | Default          |
| Shim mode                | Standard         |
| Adjust with body coil    | Off              |
| Confirm freq. adjustment | Off              |
| Assume Silicone          | Off              |
| ? Ref. amplitude 1H      | 0.000 V          |
| Adjustment Tolerance     | Auto             |
| Adjust volume            |                  |
| Position                 | L0.0 P10.9 H20.5 |
| Orientation              | Transversal      |
| Rotation                 | -180.00 deg      |
| R >> L                   | 220 mm           |
| A >> P                   | 220 mm           |
| F >> H                   | 128 mm           |

## Sequence

|                 |           |
|-----------------|-----------|
| Introduction    | On        |
| Dimension       | 2D        |
| Asymmetric echo | Off       |
| Contrasts       | 2         |
| Bandwidth       | 260 Hz/Px |
| Flow comp.      | Yes       |
| RF pulse type   | Normal    |
| Gradient mode   | Fast      |
| RF spoiling     | On        |

## SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\mls\_mb16\_400ms

TA: 4:05 PAT: Off Voxel size: 3.4×3.4×3.5 mm Rel. SNR: 1.00 USER: cmrr\_mbep2d\_bold

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                          |                   |
|--------------------------|-------------------|
| Slice group 1            |                   |
| Slices                   | 48                |
| Dist. factor             | 0 %               |
| Position                 | L1.3 A0.6 F11.1   |
| Orientation              | Transversal       |
| Phase enc. dir.          | P >> A            |
| Rotation                 | -180.00 deg       |
| Phase oversampling       | 0 %               |
| FoV read                 | 220 mm            |
| FoV phase                | 100.0 %           |
| Slice thickness          | 3.50 mm           |
| TR                       | 400 ms            |
| TE                       | 30.0 ms           |
| Multi-band accel. factor | 16                |
| Filter                   | Prescan Normalize |
| Coil elements            | HEA;HEP           |

## Contrast

|                   |           |
|-------------------|-----------|
| MTC               | Off       |
| Magn. preparation | None      |
| Flip angle        | 24 deg    |
| Fat suppr.        | Fat sat.  |
| Averaging mode    | Long term |
| Reconstruction    | Magnitude |
| Measurements      | 588       |
| Delay in TR       | 0 ms      |
| Multiple series   | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 64        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 7/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | On        |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                 |
|--------------------------|-----------------|
| Body                     | Off             |
| HEP                      | On              |
| HEA                      | On              |
| Positioning mode         | FIX             |
| Table position           | H               |
| Table position           | 0 mm            |
| MSMA                     | S - C - T       |
| Sagittal                 | R >> L          |
| Coronal                  | A >> P          |
| Transversal              | F >> H          |
| Coil Combine Mode        | Sum of Squares  |
| AutoAlign                | Head > Brain    |
| Auto Coil Select         | Default         |
| Shim mode                | Standard        |
| Adjust with body coil    | Off             |
| Confirm freq. adjustment | Off             |
| Assume Silicone          | Off             |
| ? Ref. amplitude 1H      | 0.000 V         |
| Adjustment Tolerance     | Auto            |
| Adjust volume            |                 |
| Position                 | L1.3 A0.6 F11.1 |
| Orientation              | Transversal     |
| Rotation                 | -180.00 deg     |
| R >> L                   | 220 mm          |
| A >> P                   | 220 mm          |
| F >> H                   | 168 mm          |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## BOLD

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Dynamic t-maps          | Off      |
| Starting ignore meas    | 0        |
| Ignore after transition | 0        |
| Model transition states | On       |
| Temp. highpass filter   | On       |
| Threshold               | 4.00     |
| Paradigm size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion correction       | Off      |
| Spatial filter          | Off      |

## Sequence

# SIEMENS MAGNETOM TrioTim syngo MR B17

|                          |            |
|--------------------------|------------|
| Introduction             | Off        |
| Bandwidth                | 3720 Hz/Px |
| Echo spacing             | 0.5 ms     |
| <hr/>                    |            |
| EPI factor               | 64         |
| Gradient mode            | Fast       |
| <hr/>                    |            |
| Excite pulse duration    | 4000 us    |
| Single-band images       | Off        |
| MB RF phase scramble     | Off        |
| SENSE1 coil combine      | Off        |
| Log physiology to file   | Off        |
| Invert RO/PE polarity    | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Triggering scheme        | Standard   |
| Starting ignore meas     | 0          |
| Paradigm size            | 2          |
| Multiplier               | 1          |
| Step [1]                 | 1          |
| Step [2]                 | 0          |

## SIEMENS MAGNETOM TrioTim syngo MR B17

\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\breathhold\_mb16\_400ms

TA: 6:10 PAT: Off Voxel size: 3.4×3.4×3.5 mm Rel. SNR: 1.00 USER: cmrr\_mbep2d\_bold

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                          |                   |
|--------------------------|-------------------|
| Slice group 1            |                   |
| Slices                   | 48                |
| Dist. factor             | 0 %               |
| Position                 | L1.3 A0.6 F11.1   |
| Orientation              | Transversal       |
| Phase enc. dir.          | P >> A            |
| Rotation                 | -180.00 deg       |
| Phase oversampling       | 0 %               |
| FoV read                 | 220 mm            |
| FoV phase                | 100.0 %           |
| Slice thickness          | 3.50 mm           |
| TR                       | 400 ms            |
| TE                       | 30.0 ms           |
| Multi-band accel. factor | 16                |
| Filter                   | Prescan Normalize |
| Coil elements            | HEA;HEP           |

## Contrast

|                   |           |
|-------------------|-----------|
| MTC               | Off       |
| Magn. preparation | None      |
| Flip angle        | 24 deg    |
| Fat suppr.        | Fat sat.  |
| Averaging mode    | Long term |
| Reconstruction    | Magnitude |
| Measurements      | 900       |
| Delay in TR       | 0 ms      |
| Multiple series   | Off       |

## Resolution

|                       |           |
|-----------------------|-----------|
| Base resolution       | 64        |
| Phase resolution      | 100 %     |
| Phase partial Fourier | 7/8       |
| Interpolation         | Off       |
| PAT mode              | None      |
| Matrix Coil Mode      | Auto (CP) |
| Distortion Corr.      | Off       |
| Unfiltered images     | Off       |
| Prescan Normalize     | On        |
| Raw filter            | On        |
| Elliptical filter     | Off       |
| Hamming               | Off       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Interleaved |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                 |
|--------------------------|-----------------|
| Body                     | Off             |
| HEP                      | On              |
| HEA                      | On              |
| Positioning mode         | FIX             |
| Table position           | H               |
| Table position           | 0 mm            |
| MSMA                     | S - C - T       |
| Sagittal                 | R >> L          |
| Coronal                  | A >> P          |
| Transversal              | F >> H          |
| Coil Combine Mode        | Sum of Squares  |
| AutoAlign                | Head > Brain    |
| Auto Coil Select         | Default         |
| Shim mode                | Standard        |
| Adjust with body coil    | Off             |
| Confirm freq. adjustment | Off             |
| Assume Silicone          | Off             |
| ? Ref. amplitude 1H      | 0.000 V         |
| Adjustment Tolerance     | Auto            |
| Adjust volume            |                 |
| Position                 | L1.3 A0.6 F11.1 |
| Orientation              | Transversal     |
| Rotation                 | -180.00 deg     |
| R >> L                   | 220 mm          |
| A >> P                   | 220 mm          |
| F >> H                   | 168 mm          |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
|-----------------|------|

## BOLD

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Dynamic t-maps          | Off      |
| Starting ignore meas    | 0        |
| Ignore after transition | 0        |
| Model transition states | On       |
| Temp. highpass filter   | On       |
| Threshold               | 4.00     |
| Paradigm size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion correction       | Off      |
| Spatial filter          | Off      |

## Sequence



# SIEMENS MAGNETOM TrioTim syngo MR B17

|                          |            |
|--------------------------|------------|
| Introduction             | Off        |
| Bandwidth                | 3720 Hz/Px |
| Echo spacing             | 0.5 ms     |
| <hr/>                    |            |
| EPI factor               | 64         |
| Gradient mode            | Fast       |
| <hr/>                    |            |
| Excite pulse duration    | 4000 us    |
| Single-band images       | Off        |
| MB RF phase scramble     | Off        |
| SENSE1 coil combine      | Off        |
| Log physiology to file   | Off        |
| Invert RO/PE polarity    | Off        |
| Online multi-band recon. | Online     |
| FFT scale factor         | 1.00       |
| Triggering scheme        | Standard   |
| Starting ignore meas     | 0          |
| Paradigm size            | 2          |
| Multiplier               | 1          |
| Step [1]                 | 1          |
| Step [2]                 | 0          |

## SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\bh\_pcasl\_DE\_M0

TA: 0:40 PAT: 2 Voxel size: 3.4×3.4×5.0 mm Rel. SNR: 1.00 USER: ep2d\_DE\_pcasl\_iPAT

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | On     |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 19                |
| Dist. factor       | 40 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 220 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 10000 ms          |
| TE 1               | 10.0 ms           |
| TE 2               | 25 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 90 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 1         |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 64            |
| Phase resolution      | 100 %         |
| Phase partial Fourier | 7/8           |
| Interpolation         | Off           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Separate      |
| Distortion Corr.      | Off           |
| Unfiltered images     | Off           |
| Prescan Normalize     | On            |
| Raw filter            | Off           |
| Elliptical filter     | Off           |
| Hamming               | Off           |

## Geometry

Multi-slice mode  
Series Interleaved  
Ascending

Special sat. None

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 220 mm         |
| A >> P                   | 220 mm         |
| F >> H                   | 131 mm         |

## Physio

1st Signal/Mode None

## Perf

|                 |     |
|-----------------|-----|
| GBP             | Off |
| PBP             | Off |
| TTP             | Off |
| Original images | On  |

## Sequence

|                   |             |
|-------------------|-------------|
| Introduction      | Off         |
| Contrasts         | 2           |
| Bandwidth         | 3004 Hz/Px  |
| Free echo spacing | Off         |
| Echo spacing      | 0.44 ms     |
| EPI factor        | 64          |
| RF pulse type     | Normal      |
| Gradient mode     | Fast        |
| CASL Method       | Multi-slice |
| Label Offset      | 90 mm       |
| Post Label Delay  | 500000 us   |
| Num RF Blocks     | 82          |
| RF GAP            | 360 us      |
| mean Gz x10       | 6 mT/m      |
| phi adjust        | 100 percent |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\bh\_pcasl\_DE\_1p5

TA: 6:12 PAT: 2 Voxel size: 3.4×3.4×5.0 mm Rel. SNR: 1.00 USER: ep2d\_DE\_pcasl\_iPAT

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slice group 1      |                   |
| Slices             | 19                |
| Dist. factor       | 40 %              |
| Position           | Isocenter         |
| Orientation        | Transversal       |
| Phase enc. dir.    | P >> A            |
| Rotation           | 180.00 deg        |
| Phase oversampling | 0 %               |
| FoV read           | 220 mm            |
| FoV phase          | 100.0 %           |
| Slice thickness    | 5.0 mm            |
| TR                 | 4000 ms           |
| TE 1               | 10.0 ms           |
| TE 2               | 25 ms             |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Prescan Normalize |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |           |
|-----------------|-----------|
| MTC             | Off       |
| Flip angle      | 90 deg    |
| Fat suppr.      | Fat sat.  |
| Averaging mode  | Long term |
| Reconstruction  | Magnitude |
| Measurements    | 90        |
| Delay in TR     | 0 ms      |
| Multiple series | Off       |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 64            |
| Phase resolution      | 100 %         |
| Phase partial Fourier | 7/8           |
| Interpolation         | Off           |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Separate      |
| Distortion Corr.      | Off           |
| Unfiltered images     | Off           |
| Prescan Normalize     | On            |
| Raw filter            | Off           |
| Elliptical filter     | Off           |
| Hamming               | Off           |

## Geometry

Multi-slice mode  
Series

Interleaved  
Ascending

Special sat.

None

## System

|                          |                |
|--------------------------|----------------|
| Body                     | Off            |
| HEP                      | On             |
| HEA                      | On             |
| Positioning mode         | FIX            |
| Table position           | H              |
| Table position           | 0 mm           |
| MSMA                     | S - C - T      |
| Sagittal                 | R >> L         |
| Coronal                  | A >> P         |
| Transversal              | F >> H         |
| Save uncombined          | Off            |
| Coil Combine Mode        | Sum of Squares |
| AutoAlign                | ---            |
| Auto Coil Select         | Default        |
| Shim mode                | Standard       |
| Adjust with body coil    | Off            |
| Confirm freq. adjustment | Off            |
| Assume Silicone          | Off            |
| ? Ref. amplitude 1H      | 0.000 V        |
| Adjustment Tolerance     | Auto           |
| Adjust volume            |                |
| Position                 | Isocenter      |
| Orientation              | Transversal    |
| Rotation                 | 180.00 deg     |
| R >> L                   | 220 mm         |
| A >> P                   | 220 mm         |
| F >> H                   | 131 mm         |

## Physio

1st Signal/Mode

None

## Perf

|                      |    |
|----------------------|----|
| GBP                  | On |
| PBP                  | On |
| TTP                  | On |
| Original images      | On |
| Starting ignore meas | 2  |

## Sequence

|                   |             |
|-------------------|-------------|
| Introduction      | Off         |
| Contrasts         | 2           |
| Bandwidth         | 3004 Hz/Px  |
| Free echo spacing | Off         |
| Echo spacing      | 0.44 ms     |
| EPI factor        | 64          |
| RF pulse type     | Normal      |
| Gradient mode     | Fast        |
| CASL Method       | Multi-slice |
| Label Offset      | 90 mm       |
| Post Label Delay  | 1500000 us  |
| Num RF Blocks     | 82          |
| RF GAP            | 360 us      |
| mean Gz x10       | 6 mT/m      |
| phi adjust        | 100 percent |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\PC\_3D\_30\_30\_30  
TA: 5:55 PAT: 2 Voxel size: 0.8×0.8×0.9 mm Rel. SNR: 1.00 SIEMENS: fl\_pc

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slab group 1       |                   |
| Slabs              | 1                 |
| Dist. factor       | 20 %              |
| Position           | L0.0 A17.9 H13.8  |
| Orientation        | T > C-21.9        |
| Phase enc. dir.    | R >> L            |
| Rotation           | 90.00 deg         |
| Phase oversampling | 0 %               |
| Slice oversampling | 0.0 %             |
| Slices per slab    | 160               |
| FoV read           | 200 mm            |
| FoV phase          | 75.0 %            |
| Slice thickness    | 0.90 mm           |
| TR                 | 40.95 ms          |
| TE                 | 6.21 ms           |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Elliptical filter |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |            |
|-----------------|------------|
| Flip angle      | 15 deg     |
| Averaging mode  | Short term |
| Reconstruction  | Magnitude  |
| Measurements    | 1          |
| Multiple series | Off        |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 256           |
| Phase resolution      | 100 %         |
| Slice resolution      | 50 %          |
| Phase partial Fourier | Off           |
| Interpolation         | On            |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Integrated    |
| Image Filter          | Off           |
| Distortion Corr.      | Off           |
| Prescan Normalize     | Off           |
| Normalize             | Off           |
| B1 filter             | Off           |
| Raw filter            | Off           |
| Elliptical filter     | On            |
| Mode                  | Inplane       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Sequential  |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                  |
|--------------------------|------------------|
| Body                     | Off              |
| HEP                      | On               |
| HEA                      | On               |
| Positioning mode         | REF              |
| Table position           | H                |
| Table position           | 0 mm             |
| MSMA                     | C - T - S        |
| Sagittal                 | R >> L           |
| Coronal                  | A >> P           |
| Transversal              | F >> H           |
| Coil Combine Mode        | Adaptive Combine |
| AutoAlign                | Head > Brain     |
| Auto Coil Select         | Default          |
| Shim mode                | Tune up          |
| Adjust with body coil    | Off              |
| Confirm freq. adjustment | Off              |
| Assume Silicone          | Off              |
| ? Ref. amplitude 1H      | 0.000 V          |
| Adjustment Tolerance     | Auto             |
| Adjust volume            |                  |
| Position                 | Isocenter        |
| Orientation              | Transversal      |
| Rotation                 | 0.00 deg         |
| R >> L                   | 350 mm           |
| A >> P                   | 263 mm           |
| F >> H                   | 350 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
| Segments        | 1    |

## Angio

|                      |               |
|----------------------|---------------|
| Flow mode            | Single vel.   |
| Encodings            | 3             |
| Velocity enc.        | 30 cm/s       |
| Direction 1          | Through plane |
| Direction 2          | R >> L        |
| Direction 3          | A >> P        |
| Rephased images      | On            |
| Magnitude images     | Off           |
| Magnitude sum        | On            |
| Phase images         | Off           |
| Subtract             | Off           |
| Std-Dev-Sag          | Off           |
| Std-Dev-Cor          | Off           |
| Std-Dev-Tra          | Off           |
| Std-Dev-Time         | Off           |
| MIP-Sag              | Off           |
| MIP-Cor              | Off           |
| MIP-Tra              | On            |
| MIP-Time             | Off           |
| Save original images | On            |

## Sequence

|                     |     |
|---------------------|-----|
| Introduction        | On  |
| Dimension           | 3D  |
| Elliptical scanning | Off |

## SIEMENS MAGNETOM TrioTim syngo MR B17

|                 |           |
|-----------------|-----------|
| Asymmetric echo | Weak      |
| Contrasts       | 1         |
| Bandwidth       | 300 Hz/Px |
| Flow comp.      | No        |
| <hr/>           |           |
| RF pulse type   | Normal    |
| Gradient mode   | Fast      |
| RF spoiling     | On        |

# SIEMENS MAGNETOM TrioTim syngo MR B17

\\USER\McLean Research\Frederick\NIRS\_CBV\_ToScan\PC\_3D\_75\_75\_75  
TA: 5:55 PAT: 2 Voxel size: 0.8×0.8×0.9 mm Rel. SNR: 1.00 SIEMENS: fl\_pc

## Properties

|                                               |        |
|-----------------------------------------------|--------|
| Prio Recon                                    | Off    |
| Before measurement                            |        |
| After measurement                             |        |
| Load to viewer                                | On     |
| Inline movie                                  | Off    |
| Auto store images                             | On     |
| Load to stamp segments                        | Off    |
| Load images to graphic segments               | Off    |
| Auto open inline display                      | Off    |
| Start measurement without further preparation | On     |
| Wait for user to start                        | Off    |
| Start measurements                            | single |

## Routine

|                    |                   |
|--------------------|-------------------|
| Slab group 1       |                   |
| Slabs              | 1                 |
| Dist. factor       | 20 %              |
| Position           | L0.0 A17.9 H13.8  |
| Orientation        | T > C-21.9        |
| Phase enc. dir.    | R >> L            |
| Rotation           | 90.00 deg         |
| Phase oversampling | 0 %               |
| Slice oversampling | 0.0 %             |
| Slices per slab    | 160               |
| FoV read           | 200 mm            |
| FoV phase          | 75.0 %            |
| Slice thickness    | 0.90 mm           |
| TR                 | 40.95 ms          |
| TE                 | 6.21 ms           |
| Averages           | 1                 |
| Concatenations     | 1                 |
| Filter             | Elliptical filter |
| Coil elements      | HEA;HEP           |

## Contrast

|                 |            |
|-----------------|------------|
| Flip angle      | 15 deg     |
| Averaging mode  | Short term |
| Reconstruction  | Magnitude  |
| Measurements    | 1          |
| Multiple series | Off        |

## Resolution

|                       |               |
|-----------------------|---------------|
| Base resolution       | 256           |
| Phase resolution      | 100 %         |
| Slice resolution      | 50 %          |
| Phase partial Fourier | Off           |
| Interpolation         | On            |
| PAT mode              | GRAPPA        |
| Accel. factor PE      | 2             |
| Ref. lines PE         | 24            |
| Matrix Coil Mode      | Auto (Triple) |
| Reference scan mode   | Integrated    |
| Image Filter          | Off           |
| Distortion Corr.      | Off           |
| Prescan Normalize     | Off           |
| Normalize             | Off           |
| B1 filter             | Off           |
| Raw filter            | Off           |
| Elliptical filter     | On            |
| Mode                  | Inplane       |

## Geometry

|                  |             |
|------------------|-------------|
| Multi-slice mode | Sequential  |
| Series           | Interleaved |
| Special sat.     | None        |

## System

|                          |                  |
|--------------------------|------------------|
| Body                     | Off              |
| HEP                      | On               |
| HEA                      | On               |
| Positioning mode         | FIX              |
| Table position           | H                |
| Table position           | 0 mm             |
| MSMA                     | C - T - S        |
| Sagittal                 | R >> L           |
| Coronal                  | A >> P           |
| Transversal              | F >> H           |
| Coil Combine Mode        | Adaptive Combine |
| AutoAlign                | Head > Brain     |
| Auto Coil Select         | Default          |
| Shim mode                | Tune up          |
| Adjust with body coil    | Off              |
| Confirm freq. adjustment | Off              |
| Assume Silicone          | Off              |
| ? Ref. amplitude 1H      | 0.000 V          |
| Adjustment Tolerance     | Auto             |
| Adjust volume            |                  |
| Position                 | Isocenter        |
| Orientation              | Transversal      |
| Rotation                 | 0.00 deg         |
| R >> L                   | 350 mm           |
| A >> P                   | 263 mm           |
| F >> H                   | 350 mm           |

## Physio

|                 |      |
|-----------------|------|
| 1st Signal/Mode | None |
| Segments        | 1    |

## Angio

|                      |               |
|----------------------|---------------|
| Flow mode            | Single vel.   |
| Encodings            | 3             |
| Velocity enc.        | 75 cm/s       |
| Direction 1          | Through plane |
| Direction 2          | R >> L        |
| Direction 3          | A >> P        |
| Rephased images      | On            |
| Magnitude images     | Off           |
| Magnitude sum        | On            |
| Phase images         | Off           |
| Subtract             | Off           |
| Std-Dev-Sag          | Off           |
| Std-Dev-Cor          | Off           |
| Std-Dev-Tra          | Off           |
| Std-Dev-Time         | Off           |
| MIP-Sag              | Off           |
| MIP-Cor              | Off           |
| MIP-Tra              | On            |
| MIP-Time             | Off           |
| Save original images | On            |

## Sequence

|                     |     |
|---------------------|-----|
| Introduction        | On  |
| Dimension           | 3D  |
| Elliptical scanning | Off |

## SIEMENS MAGNETOM TrioTim syngo MR B17

|                 |           |
|-----------------|-----------|
| Asymmetric echo | Weak      |
| Contrasts       | 1         |
| Bandwidth       | 300 Hz/Px |
| Flow comp.      | No        |
| <hr/>           |           |
| RF pulse type   | Normal    |
| Gradient mode   | Fast      |
| RF spoiling     | On        |

## Table of contents

\\USER

McLean Research  
Frederick

```

NIRS_CBV_ToScan
  Optimized for 32 channel!
  Head First Supine
  Begin NIRS recording
  localizer
  AAHScout
  me_mprage
  -----Finish VA Brain (if needed)-----
  DOUBLE ECHO TSE low sar
  Flair axial low SAR
  slice_positioning
  -----VASO_baseline-----
  ep2d_se_VASO
  VASO_flipback_464_550
  VASO_flipback_614_788
  VASO_flipback_764_1085
  VASO_flipback_914_1492
  VASO_flipback_1066_2190
  -----CBV/NIRS_baseline-----
  restingstate
  -----ASL_baseline-----
  Call Gordana x2957 for 7 min warning
  pcal_DE_M0
  Make sure the next scan uses BC
  pcal_DE_B0
  pcal_DE_0p5
  pcal_DE_1p0
  pcal_DE_1p5
  pcal_DE_2p0
  pcal_DE_2p5
  -----DSC-MRI-----
  Start scan at same time as injector
  Start both with 3, 2, 1, start
  dsc_mri
  gre_field_mapping
  -----MLS-----
  Open fmri_checkerflash.py
  Load mls_6bit_3pertr.fstim
  mls_mb16_400ms
  -----CBV/NIRS_breathhold-----
  Open fmri_multistim.py
  Load breathe.sstim
  breathhold_mb16_400ms
  -----ASL_breathhold-----
  Call Lia for 7 min warning
  Load breathe.sstim
  bh_pcal_DE_M0
  bh_pcal_DE_1p5
  -----Vessel maps-----
  PC_3D_30_30_30
  PC_3D_75_75_75

```